Lecture 23
Significance Tests

Review

1 What p* do we use to plan n when there is no earlier guess?

p* = 0.5: it makes p*(1 − p*) largest, so n is big enough whatever p is.

2 What is the null hypothesis H_0?

“No effect”: nothing interesting is happening, e.g. they are just guessing.

3 What is the alternative hypothesis H_A?

“An effect”: what we suspect is true, e.g. they can taste the difference.

Can you taste the difference?

Your roommate claims they can tell Coke from Pepsi. In a blind test, you pour one of the two at random, 8 times, and they name each cup.

8 of 8 cups named correctly

(1) Can they really tell, or were they just lucky? How could we decide?
(2) If they were only guessing, how likely is 8 for 8?
(3) What are the two ways to explain such a small probability?

(2) $(1/2)^8 = 1/256 \approx 0.004$
(3) Either something rare happened by chance, or "just guessing" is wrong
How rare is rare enough is our choice; by convention, below 0.05

Why test a hypothesis?

hypothesis

a claim about a population parameter (p or μ), e.g. p = 0.5: "they are just guessing"

purpose of a significance test

to decide whether the data give convincing evidence against a hypothesis

the logic

assume the null hypothesis is true, then ask how unlikely our data would be under that hypothesis

Significance tests

statistically significant

decidedly not due to ordinary variation: not chance, not a coincidence

significance test

uses data to decide whether a result is statistically significant

null hypothesis H_0

"no effect": nothing interesting is happening, e.g. they are just guessing

alternative hypothesis H_A

"an effect": what we suspect, e.g. they can taste the difference

Writing the hypotheses

θ and θ_0 ("theta" and "theta-naught")

θ is the parameter (p or μ); θ_0 is the value H_0 claims, e.g. $p_0 = 0.5$

$H_0: \theta = \theta_0$

H_0 always has the = sign, and is about a parameter, never $\hat{p}$ or $\bar{x}$

$H_A: \theta > \theta_0$, $\theta < \theta_0$ or $\theta \neq \theta_0$

upper-tailed, lower-tailed (one-sided) or two-sided; set before seeing data

Example: stating hypotheses

For each question, write H_A first (the suspicion), then H_0 (no effect). Give θ_0 and say which tail.

p = a population proportion; μ = a population mean

(a) Your roommate names Coke or Pepsi. Do they do better than guessing?
(b) A coin is spun on a table. Does it favor heads or tails?
(c) A chip bag says 28 g. Do the bags hold less than that on average?

(a) H_A: $p > 0.5$ H_0: $p = 0.5$ $p_0 = 0.5$, upper-tailed (p = chance of a right cup)
(b) H_A: $p \neq 0.5$ H_0: $p = 0.5$ $p_0 = 0.5$, two-sided (p = chance of heads)
(c) H_A: $\mu < 28$ g H_0: $\mu = 28$ g $\mu_0 = 28$ g, lower-tailed

Conditional probability

conditional probability P(A | B)

the probability that A happens, given that B is true; the bar | is read "given"

in a significance test

every probability is worked out given H_0 is true: P(8 of 8 | H_0 true) = $0.5^8 \approx 0.004$

Test statistics and P-values

test statistic z_{obs}

z_{obs} = (estimate − θ_0) / SE: how many SEs the estimate is from θ_0

P-value

the probability of observing a value as or more extreme than the test statistic, when H_0 is true
upper-tailed: $P(Z \geq z_{obs} \mid H_0 \text{ true})$

significance level α

the cutoff, chosen before collecting data; usually $\alpha = 0.05$

the decision

P-value < α: reject H_0 (significant); P-value ≥ α: do not reject H_0

Example: taste-test P-values

A second friend makes the same claim and takes a 6-cup test. Test H_0: $p = 0.5$ against H_A: $p > 0.5$ at $\alpha = 0.05$.

roommate: 8 of 8 correct; friend: 4 of 6 correct

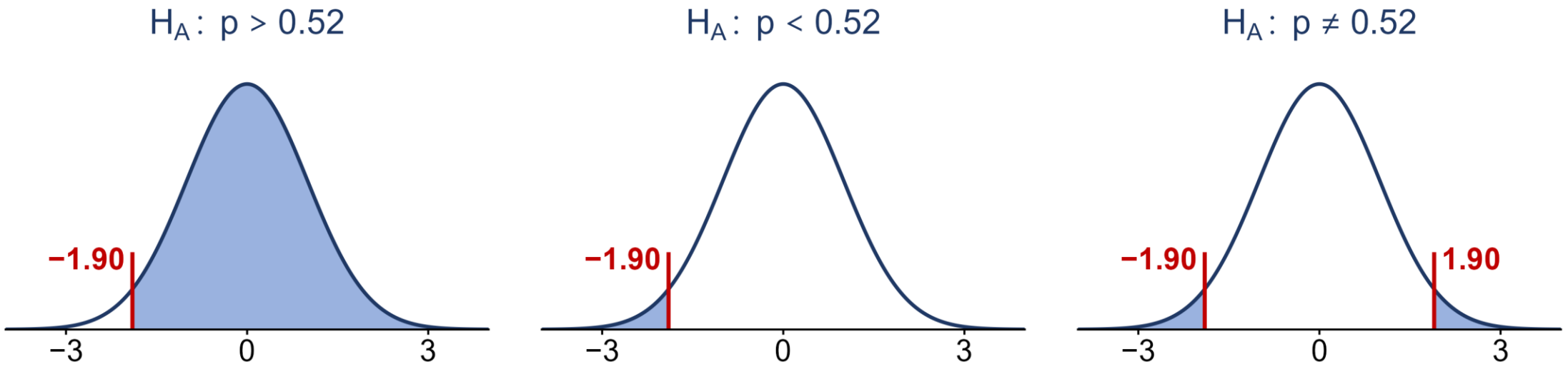
(a) If H_0 is true, what is the P-value for each?
(b) What is the decision for each?
(c) If 256 people took the 8-cup test by pure guessing, how many would go 8 for 8?

(a) roommate: $P(8 \text{ of } 8 \mid H_0 \text{ true}) = 0.5^8 \approx 0.004$
friend: $P(4 \text{ or more of } 6 \mid H_0 \text{ true}) = 22/64 \approx 0.34$
(b) roommate: $0.004 < 0.05$, reject H_0; friend: $0.34 \geq 0.05$, do not reject H_0
(c) About 1: a small P-value means rare luck or a false H_0, and we can't be sure which

Which tail is the P-value?

Polls put Candidate X at 52%. Our poll of 1,000 likely voters finds 49.0%, so testing H_0: p = 0.52 gives $z_{obs} = -1.90$.

What is the P-value under each H_A? Which ones reject H_0 at α = 0.05?



upper: P(Z ≥ −1.90) = 0.971 lower: P(Z ≤ −1.90) = 0.029 two: 2 × 0.029 = 0.057

Only the lower-tailed test rejects H_0: H_A must be set before seeing the data

Testing a population proportion

when to use it

a categorical variable, and a claim H_0: $p = p_0$ about the proportion of successes

assumptions

a random sample, and $np_0 \geq 15$ and $n(1 - p_0) \geq 15$

test statistic

$z_{obs} = (\hat{p} - p_0) / \sqrt{(p_0(1 - p_0) / n)}$: the SE uses p_0, because we assume H_0 is true

P-value

upper: $P(Z \geq z_{obs} \mid H_0 \text{ true})$; lower: $P(Z \leq z_{obs} \mid H_0 \text{ true})$; two-sided: double the tail

Example: cell phones and brain cancer

In a study of 420,019 cell phone users, 172 developed brain cancer. For people who do not use cell phones, the rate is 0.034%. Do cell phone users get brain cancer at a higher rate? (OpenStax, Example 9.21)

$x = 172$; $n = 420{,}019$; $p_0 = 0.00034$; $\alpha = 0.005$

(a) What are H_0 and H_A? Which tail? Are the assumptions met?
(b) What are $\hat{p}$, the standard error and z_{obs}?
(c) What is the P-value? What is the decision at $\alpha = 0.005$?
(d) Why would we choose such a small α here?

(a) $H_0: p = 0.00034$ $H_A: p > 0.00034$, upper-tailed; $np_0 = 142.8 \geq 15$, $n(1 - p_0) \geq 15$
(b) $\hat{p} = 172 / 420{,}019 \approx 0.0004095$; $SE = \sqrt{p_0(1 - p_0) / n} \approx 0.00002845$
$z_{obs} = (0.0004095 - 0.00034) / 0.00002845 \approx 2.44$
(c) $P(Z \geq 2.44 \mid H_0 \text{ true}) = 1 - 0.9927 = 0.0073 \geq 0.005$: do not reject H_0
(d) Rejecting a true H_0 would wrongly blame phones for cancer, so we ask for strong evidence

Assumptions for a population proportion

both

a random sample; each individual is a success or a failure

confidence interval for p

$n\hat{p} > 5$ and $n(1 - \hat{p}) > 5$: checked with $\hat{p}$, because p is unknown

significance test for p

$np_0 \geq 15$ and $n(1 - p_0) \geq 15$: checked with p_0, because we assume H_0 is true

why they matter

they make the sampling distribution of $\hat{p}$ close to normal, so z-table areas are right

The five steps of a significance test

1. assumptions, 2. hypotheses

a random sample (plus size or shape checks); H_0: $\theta = \theta_0$ and H_A

3. test statistic

z_{obs} = (estimate – θ_0) / SE, computed as if H_0 were true

4. P-value

the area beyond z_{obs} in the tail or tails H_A points to, if H_0 were true

5. decision

P-value < α: reject H_0; otherwise do not reject; significant ≠ important

Practice: is breakfast-skipping up?

Last year 30% of students skipped breakfast. This fall, campus dining asks a random sample of 400 students, and 136 say they skip it. Has the proportion gone up?

$n = 400$; 136 skip breakfast; $p_0 = 0.30$; $\alpha = 0.05$

(a) Are the assumptions met?
(b) What are H_0 and H_A? Which tail?
(c) What are $\hat{p}$, the standard error and z_{obs}?
(d) What is the P-value? What is the decision?

(a) a random sample; $np_0 = 400(0.30) = 120 \geq 15$ and $n(1 - p_0) = 280 \geq 15$
(b) H_0: $p = 0.30$ H_A: $p > 0.30$; upper-tailed
(c) $\hat{p} = 136 / 400 = 0.34$; $SE = \sqrt{(0.30 \times 0.70 / 400)} \approx 0.0229$; $z_{obs} = 0.04 / 0.0229 \approx 1.75$
(d) $P(Z \geq 1.75 \mid H_0 \text{ true}) = 1 - 0.9599 = 0.0401 < 0.05$: reject H_0